

**UNIT IV****6 Hours****CATALYSIS***LP I : Introduction to catalysis***INTRODUCTION TO CATALYSIS**

Catalysis touches everyone's life from the catalytic converter in your vehicle, to how you clean your contact lenses, to creating your plastic water bottle. Though catalysts have been used by humans since ancient times, such as to produce alcohol through the fermentation of sugars, the systematic study of catalysis did not begin until the 1700s. In the late 1700s and early 1800s, Elizabeth Fulhame provided a theoretical foundation for the study of catalysts. In 1835, the word "catalysis" was coined by chemist Jöns Jakob Berzelius. Throughout the eighteenth and nineteenth centuries, experiments involving various metals provided evidence that these materials could be both selective and specific catalysts. The work of Paul Sabatier in the late 1800s and early 1900s led to the discovery of many metal catalysts, particularly nickel and platinum group metals. These remain some of the most important catalysts for industrial applications to date for a variety of applications such as crude oil refining and reducing car emissions.

**DEFINITION**

*The process by which a substance speeds up a chemical reaction without being consumed or altered in the process is catalysis (or) Catalysis is the process of adding a catalyst to facilitate a reaction.*

***Catalyst***

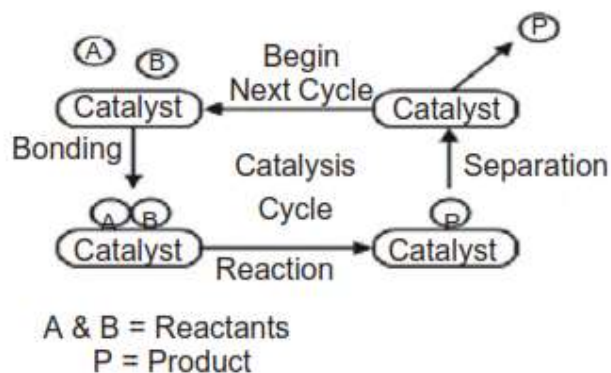
*A catalyst is a substance that speeds up a chemical reaction, or lowers the temperature or pressure needed to start one, without itself being consumed during the reaction.*

**(or)**

*A catalyst is a substance that decreases the activation energy of a chemical reaction without itself being changed at the end of the chemical reaction*

**(or)**

*It is the small amount of substance which alters the velocity of reaction without altering in any change in mass and composition at the end of the reaction.*



Catalysis cycle

### Importance of Catalysis

Catalysts have many industrial uses, from the production of food products, biofuels, pharmaceuticals, plastics, and chemicals. Most commercially produced chemicals are made with the help of catalysts. Some notable reactions enhanced by catalysis include production of sulphuric acid used for rust removal as well as a precursor for phosphoric acid, synthesis of ammonia, petroleum refining, and limiting of vehicle emissions. Because of how integral catalysis is to many industrial processes, scientific research to improve catalysis will benefit many areas of our lives.

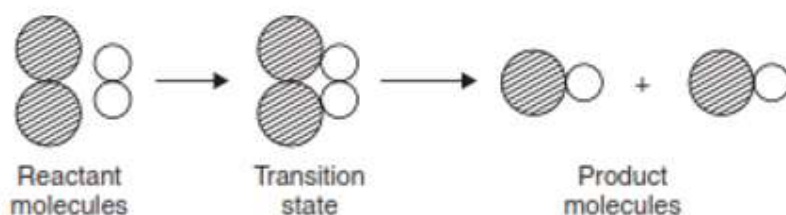
During a chemical reaction, the bonds between the atoms in molecules are broken, rearranged, and rebuilt, recombining the atoms into new molecules. Catalysts make this process more efficient by lowering the **activation energy**, which is the energy barrier that must be surmounted for a chemical reaction to occur. As a result, catalysts make it easier for atoms to break and form chemical bonds to produce new combinations and new substances.

Using catalysts leads to faster, more energy-efficient chemical reactions. Catalysts also have a key property called **selectivity**, by which they can direct a reaction to increase the amount of desired product and reduce the amount of unwanted by products. They can produce entirely new materials with entirely new potential uses.

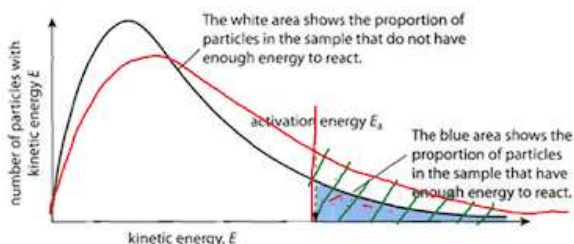
Over the past several decades, scientists have developed increasingly specialized catalysts for essential real-world applications. In particular, powerful catalysts have transformed the chemical industry. These advances have led to biodegradable plastics, new pharmaceuticals, and environmentally safer fuels and fertilizers.

### Arrhenius theory of activation energy

According to collision theory: the molecules of reactants collide to give products. Schematically it is represented as follows:



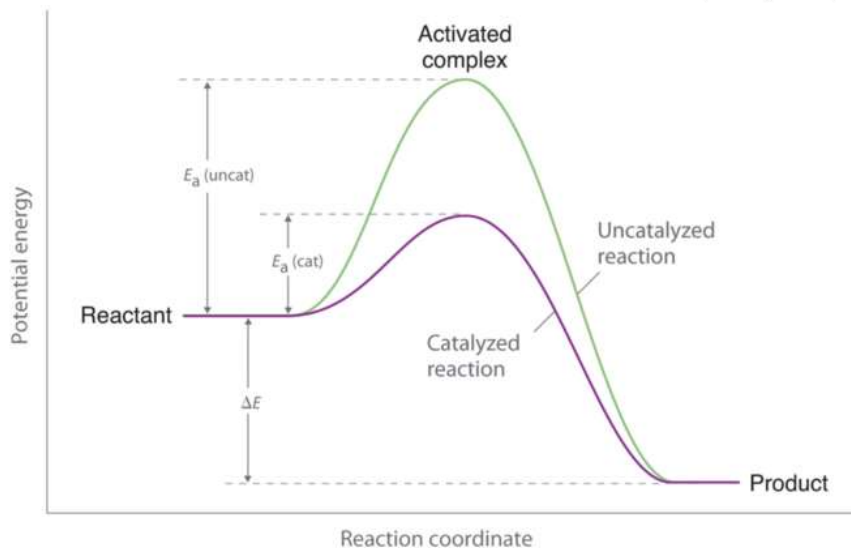
All molecules possess a certain minimum amount of energy. The energy can be in the form of kinetic energy or potential energy. When molecules collide, the kinetic energy of the molecules can be used to stretch, bend, and ultimately break bonds, leading to chemical reactions. If molecules move too slowly with little kinetic energy, or collide with improper orientation, they do not react and simply bounce off each other. However, if the molecules are moving fast enough with a proper collision orientation, such that the kinetic energy upon collision is greater than the minimum energy barrier, then a reaction occurs. The minimum energy requirement that must be met for a chemical reaction to occur is called the activation energy,  $E_a$ .



In Greek mythology Sisyphus was punished by being forced roll an immense boulder up a hill, only to watch it roll back down, and to repeat this action forever. If this were a chemical reaction, then it would never be observed, since the reactants must overcome the energy barrier to get to the other side (products).

### Role of a catalyst

A reaction will not take place unless the particles collide with certain minimum energy called the activation energy of the reaction. Activation energy is the minimum energy required to make a reaction occur. The frequency of collisions determines the reaction rate. A catalyzed pathway has lower activation energy.

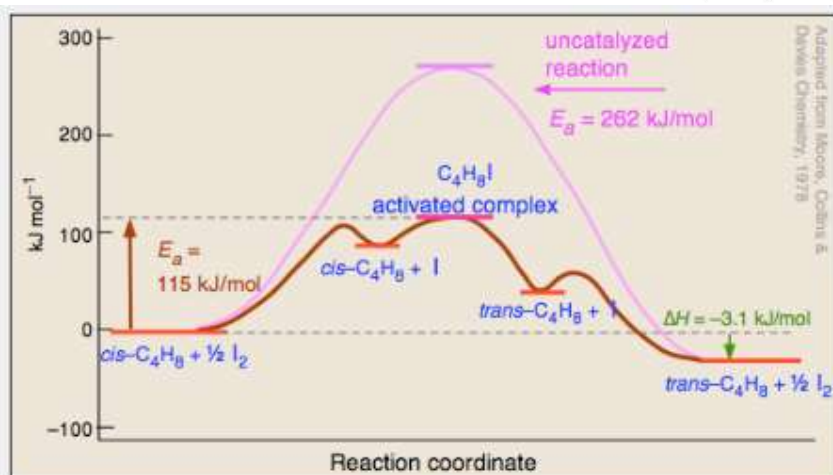


The minimum amount of energy required to attain transition state is called energy of activation,  $E_a$ . Catalyst lowers the energy of activation ( $E_a$ ) of a reaction.

Catalysts are chemicals that speed up the rate of a chemical reaction without consuming energy in the process. As a result, a catalyst does not appear in the overall stoichiometry of the reaction it catalyzes, but it does have to exist in at least one of the elementary reactions in the catalyzed reaction's mechanism. Although the catalyzed pathway has a lower  $E_a$ , the net change in energy (the difference between the energy of the reactants and the energy of the products) is unaffected by the presence of a catalyst, see the image. Nonetheless, at the same temperature, the reaction rate of a catalyzed reaction is faster than that of an un-catalyzed reaction due to its lower  $E_a$ .

Catalysts do not just reduce the energy barrier, but induced a completely different reaction pathway typically with multiple energy barriers that must be overcome. For example:

- ✓ Lowering the energy of the transition state, thus lowering the activation energy, and/or
- ✓ Changing the mechanism of the reaction. This also changes the nature (and energy) of the transition state.



## The Iodine-catalyzed cis-trans isomerization

Because a catalyst lowers the energy barrier, it enhances the rate of forward and reverse reactions by the same amount in an equilibrium reaction.

### Questions

- Draw a distribution curve showing how the proportion of molecules able to react increases when a catalyst lowers the activation energy.
- “A catalyst doesn't affect the final state of equilibrium, although it shortens the time required to establish the equilibrium.” - Justify
- Why do we prefer finely divided catalysts?